

# DISKO 1

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

## Challenge Description

---

Can you find the flag in this disk image?

## Hints

- Maybe **Strings** could help? If only there was a way to do that?
- 

## Tools Used

---

- **7z:** Decompression tool for extracting gzipped files
  - **file:** Utility to identify file types and filesystem information
  - **strings:** Command-line tool to extract readable ASCII sequences from binary files
  - **grep:** Text search utility for filtering output
- 

## Complete Solution

---

### Step 1: Download and Extract

Download the disk image named `disko-1.dd.gz` from the challenge page.

Extract the compressed disk image:

```
7z x disko-1.dd.gz
```

This will decompress the gzipped file and create the raw disk image `disko-1.dd`.

### Step 2: Basic File Inspection

Verify the file type and understand its structure:

```
file disko-1.dd
```

### Expected output:

```
disko-1.dd: DOS/MBR boot sector, code offset 0x58+2, OEM-ID "mkfs.fat", Media descriptor 0xf8, sectors/track 32, heads 8, sectors 102400 (volumes > 32 MB), FAT (32 bit), sectors/FAT 788, serial number 0x241a4420, unlabeled
```

The output confirms this is a FAT32 filesystem. This is a common filesystem format used on USB drives and memory cards.

### Step 3: Search for Visible Text

Disk images often contain strings embedded directly in the data. Extract all readable character sequences and search for the flag:

```
strings disko-1.dd | grep -i "picoCTF"
```

The `strings` command scans the binary file and extracts all printable ASCII sequences. This is often effective for finding flags hidden in filesystem data.

### ✅ Final Result

The flag is found directly in the disk image:

```
picoCTF{<REDACTED>}
```

---

## Key Concepts

- **Disk Images:** Raw copies of disk or filesystem data that can be analyzed like any file.
- **FAT32 Filesystem:** Legacy filesystem format commonly used on removable media.
- **Strings Extraction:** A powerful technique for finding readable text embedded in binary files.
- **Binary Analysis:** Sometimes flags are directly visible as strings within binary data without needing specialized tools.